

# Nidhi Dhameliya

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## PROFESSIONAL SUMMARY

AI/ML enthusiast with experience in Machine Learning, Deep Learning, LLMs, and RAG. Skilled in Python, TensorFlow, and PyTorch, with a passion for building AI-driven solutions to real-world problems.

## EDUCATION

### Rashtriya Raksha University

*M.Tech in Data Science and Machine Learning*

- CGPA: 8.59, Currently in 3rd semester

Gandhinagar, Gujarat

2025 – 2027

### Government Engineering College (GEC) Bhavnagar

*B.E. in Computer Engineering*

- CGPA: 8.89

Bhavnagar, Gujarat

2021 – 2025

## WORK EXPERIENCE

### INFOLABZ IT SERVICES PVT LTD

*Data Analytics & Machine Learning Intern*

- Learned core concepts of machine learning, data analysis, and data preprocessing, while gaining hands-on experience with APIs for model integration, data retrieval, and workflow automation.

Ahmedabad, Gujarat

July 2024

### DATAWIND DIGITAL SERVICES

*Data Analytics Intern*

- Worked on advanced data wrangling, feature engineering, time series forecasting, and real-time dashboard creation using tools like Pandas, Scikit-learn, Power BI.
- Gained hands-on experience in building interactive dashboards, automating reports, and analyzing trends for data-driven insights.

Jan 2025

## PROJECTS

### DataDesk – Browser-Based SQL Practice Platform | *React, WebAssembly, SQLite, Supabase, Groq* | [GitHub](#) | [Live](#)

- Engineered a browser-based SQL platform featuring a WebAssembly-powered SQLite engine, Monaco Editor integration, query validation, execution-plan analysis, and interactive schema visualization.
- Integrated Supabase Auth, PostgreSQL-backed progress tracking, CSV/SQLite dataset import, and Groq LLM APIs for intelligent SQL feedback and interview question generation.

### RFInsight – Wireless Signal Classification | *Python, PyTorch, ResNet50, EfficientNet, ViT* | [GitHub](#)

- Built a vision-based RF signal classification framework by converting wireless signals into spectrogram images and applying CNN and Transformer models for automated feature learning.
- Integrated transfer learning, class imbalance handling (LDAM), and Grad-CAM explainability to enhance classification performance and model interpretability across 21 RF signal classes.

### AI Research Assistant | *Next.js, FastAPI, ChromaDB, Groq, LangChain* | [GitHub](#) | [Live](#)

- Built an end-to-end RAG platform supporting multi-format document ingestion, GitHub repository indexing, hybrid retrieval (ChromaDB + BM25 + RRF + MMR), and source-cited question answering.
- Implemented a FastAPI backend with streaming LLM responses (SSE), local embedding generation, JWT authentication, SQLite-based chat history and analytics, and a Next.js frontend for document comparison and knowledge extraction.

### Climate Forecasting System | *ConvLSTM, CNN-LSTM, Transformer, PyTorch, ERA5, Xarray, Streamlit* | [GitHub](#) | [Live](#)

- Developed an end-to-end spatiotemporal climate forecasting pipeline using the ERA5 Climate Reanalysis dataset, including automated preprocessing, sliding-window sequence generation, and next-day temperature prediction over gridded climate data.
- Implemented and benchmarked ConvLSTM, CNN-LSTM, and Transformer models in PyTorch, achieving high-accuracy regional temperature forecasting with evaluation metrics, experiment tracking, and an interactive Streamlit dashboard for visualization and inference.

### Hospitality Domain Analytics Dashboard | *Power BI, DAX, Power Query, Excel* | [GitHub](#)

- Built an interactive Power BI dashboard by integrating and analyzing 5 datasets containing hotel booking, occupancy, and revenue information.

- Tracked occupancy rate, revenue, booking trends, ADR, and RevPAR using DAX for custom KPI calculations and Power Query for data cleaning and transformation.

#### **Mental Health Trend Analysis using Social Media Data** | *Python, NLTK, LDA, Scikit-learn, Power BI* | [GitHub](#)

- Analyzed social media posts to assess public sentiment and identify emerging mental health trends using VADER sentiment analysis and topic modeling techniques.
- Extracted key discussion themes with CountVectorizer and LDA, and explored correlations between mental health conversations and real-world events.
- Forecasted future mental health trends using ARIMA and developed interactive dashboards in Power BI for visualization and insights.

## SKILLS

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**Programming Languages:** Python, JavaScript

**Frontend:** React.js, Next.js, HTML, CSS

**Backend & APIs:** FastAPI, Node.js, REST APIs

**Databases & Vector Stores:** PostgreSQL, MySQL, MongoDB, ChromaDB

**ML / DL Frameworks:** TensorFlow, PyTorch, Scikit-learn

**NLP / LLMs:** LangChain, Hugging Face, Transformers, OpenAI API, RAG, Text Embeddings, Whisper

**Computer Vision:** YOLOv8, OpenCV

**Data Analysis & Visualization:** Pandas, NumPy, Power BI, Tableau, ETL, Data Cleaning & Preprocessing

**Deployment & Tools:** Docker, Git, GitHub, Streamlit

## CERTIFICATIONS

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Introduction to Large Language Models — Simplilearn, 2026

Intro to Deep Learning — Kaggle, 2026

Intro to Machine Learning — Kaggle, 2026

Artificial Intelligence Fundamentals — IBM, 2026

Complete Data Analyst Bootcamp from Basics to Advanced — Udemy, 2026

Data Visualisation: Empowering Business with Effective Insights Job Simulation — Tata Group / Forage, 2024